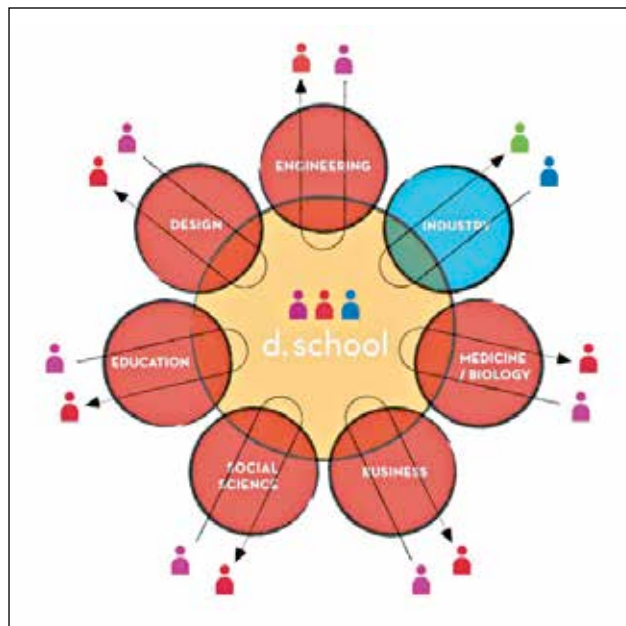


iii. **Tangibility Rule:** Offer a tangible solution to facilitate communication.

2. ACHIEVE THE A-HA!

The "A-Ha Moment" is the moment where there's suddenly a clear forward path. It's the point in the cycle where synthesis and divergent thinking, analysis and convergent thinking, and the nature of the problem all come together and an appropriate resolution has been captured. Prior to this point, the process may seem hazy and inexact. At this point, however, the path forward is so obvious that in retrospect it seems odd that it took so long to recognise it. After this point, the focus becomes clear as the final product is constructed.



The domain of Design Thinking

3. AVOID THE FEAR OF FAILURE

Fear of failure or criticism may prevent someone from even beginning to apply methods and processes to achieve their goals. Both have psychological effects which divert your mind to doubts of self worth or anxieties of "will it be good enough," or procrastination, instead of focusing on the solutions.

4. AVOID RESISTANCE

Resistance can inhibit design thinking by reprioritising the main goal and shifting efforts to other tasks which may need to be done.

5. KICK THE DEVIL'S ADVOCATE

The Devil's Advocate is an embodiment of negative criticism. Devil's advocates kill projects by shifting the focus from potential solutions to hypercritical issues with ambiguous effects.

6. USE THE "FIVE WHYS"

Design thinking includes interviewing people, creating user profiles, looking at other existing solutions, creating prototypes, mind-mapping and situational analysis. Most of this is achieved by asking questions like the "Five Whys".

7. USE VISUAL ANALOGY

Ill-defined or wicked problems often contain higher-order and obscure relationships. Design thinking can address these through the use of analogies. An understanding of the expected

results, all for a lack of domain-related knowledge, may be achieved by correlating different internal representations, such as images, to develop an understanding of the obscure elements of the situation.

STANFORD'S D.SCHOOL

Unlike analytical thinking, design thinking is a process which includes the "build up" of ideas, with few or no limits on breadth during a brainstorming phase. This helps reduce fear of failure in the participant(s) and encourages input and participation from a wide variety of sources in the ideation phases. The phrase "outside the box" thinking has been coined to describe one goal of the brainstorming phase. Such thinking is encouraged since it can aid in the discovery of hidden elements and

ambiguities in the situation as well as discovery of potentially faulty assumptions. While the stages are simple enough, the adaptive expertise suggests that choosing the right inflection points and appropriate next stage is a high order Intellectual activity that requires practice.

In organisation and management theory, design thinking forms part of the Architecture/Design/Anthropology (A/D/A) paradigm, which characterises innovative, human-centered enterprises. This paradigm also focuses on a collaborative and iterative style of work and mode of thinking, compared to practices associated with the more traditional Mathematics/Economics/Psychology (M/E/P) management paradigm. Companies that integrate the principles of design thinking in their innovation processes often

share a certain mindset or strive to cultivate a more creative and human-centered company culture.

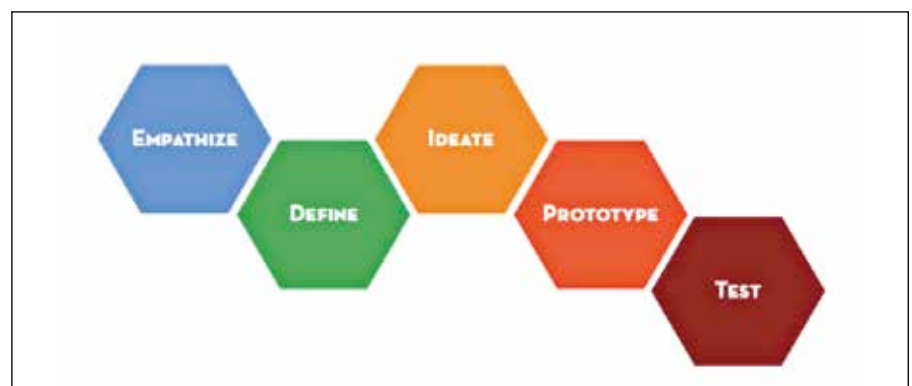
In 2005, Stanford University began teaching engineering students "Design Thinking" as a formal method in the "d.school". In a nutshell, a d.thinking project guides you through a series of phases that oscillate between divergent and convergent thinking. The phases of a d.thinking project are as follows:

1. EMPATHISE WITH END USERS

To create meaningful innovations, you need to know your users and care about their lives.

What?

Empathy is the centerpiece of a human-cen-



The phases involved in d.thinking